

1. 设概率为 P

$$80\% \cdot T_w + 10 \times 80\% \cdot T_w \times P \leq T_w$$

$$\therefore P \leq 0.025 \quad \text{即 } P \leq 2.5\%$$

2. $\sigma_{tp} = |C_L [\frac{1}{I_d} - I_d^{-2} g_m V_{DD}] \sigma_{V_{DD}}|$

$$t_p = C_L V_{DD} / I_{d_{ave}}$$

$$\therefore \frac{\sigma_{tp}}{t_p} = |(1 - I_d^{-1} g_m V_{DD}) \cdot \frac{\sigma_{V_{DD}}}{V_{DD}}|$$

(1) $I_d = k(\frac{W}{L})(V_{DD} - V_T)^2$, $g_m = \frac{\partial I_d}{\partial V_{DD}} = 2k(V_{DD} - V_T)$

$$\therefore \frac{\sigma_{tp}}{t_p} = |1 - \frac{2V_{DD}}{V_{DD} - V_T}| \cdot \frac{\sigma_{V_{DD}}}{V_{DD}} = |1 - \frac{2 \times 0.5}{0.5 - 0.4}| \cdot 10\% = 90\%$$

(2) $I_d = I_0(\frac{W}{L}) \exp(\frac{V_{DD} - V_T}{n\phi_T})$, $g_m = \frac{1}{n\phi_T} I_d$

$$\therefore \frac{\sigma_{tp}}{t_p} = |1 - \frac{V_{DD}}{n\phi_T}| \cdot \frac{\sigma_{V_{DD}}}{V_{DD}} = |1 - \frac{0.5}{1.5 \times 0.026}| \cdot 10\% = 66.9\%$$